

**Time:** 5 minutes**Outcome:** Use marginal productivity to analyze labor demand, wages, and employment.**Difficulty:** ●●○ Intermediate

## Factor Markets & Labor Demand

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### THE CORE IDEA

Labor demand is derived from the value workers add to output. Marginal product of labor is the extra output from one more worker. For a price-taking seller,  $VMP = \text{output price} \times MPL$ . A profit-maximizing firm hires labor up to the point where VMP equals the wage.

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### HOW TO RECOGNIZE IT

- A wage change causes movement along labor demand; productivity or output-price changes shift labor demand.
- Higher product demand, output price, or worker productivity shifts labor demand right.
- Worker preferences, population, migration, training, and alternative jobs can shift labor supply.
- Competitive wage and employment occur where market labor demand intersects market labor supply.

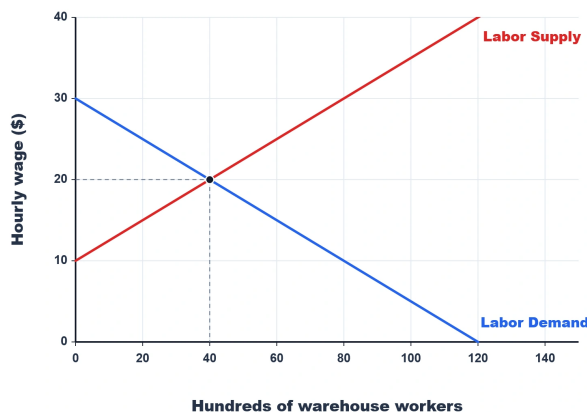
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### WATCH OUT

Do not confuse MPL with VMP. MPL is extra physical output; VMP converts that output into dollars by multiplying by the output price.

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### WORKED EXAMPLE: COMPETITIVE LABOR-MARKET EQUILIBRIUM



Warehouse labor demand and labor supply intersect at an hourly wage of \$20 and 40 hundreds of workers. Because the horizontal axis is measured in hundreds, equilibrium employment is 4,000 workers. Below \$20, firms demand more labor than workers supply; above \$20, labor supplied exceeds labor demanded.

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### CHECK YOURSELF

If the output price rises while worker productivity is unchanged, what happens to labor demand?

**READY?****Return to the game and master this concept.**